Functions in the form (ax + b) to a power

1 Find the following indefinite integrals, using C as the constant of integration:

a $\int (x-5)^4 \, dx$

b $\int \frac{1}{(x+6)^3} dx$

c $\int 6(x+2)^2 \, dx$

d $\int \frac{4}{(x-5)^3} dx$

e $\int (2x+3)^3 \, dx$

f $\int (4x-3)^{-4} \, dx$

g $\int \frac{1}{(2x-3)^4} dx$

h $\int \frac{1}{(4x+3)^2} dx$

i $\int (2x-5)^{\frac{1}{2}} \, dx$

j $\int (4x+5)^{-\frac{1}{2}} \, dx$

k $\int (4x+5)^{\frac{5}{4}} \, dx$

l $\int (1-x)^6 \, dx$

m $\int 4(5-x)^3 \, dx$

n $\int (4-3x)^6 \, dx$

o $\int 4(2x+5)^3 \, dx$

p $\int 4(2-3x)^3 \, dx$

q $\int (4-3x)^{-5} \, dx$

r $\int \frac{5}{(5-4x)^3} dx$

s $\int \sqrt{7-4x} \, dx$

t $\int 3\sqrt{2x+3} \, dx$

u $\int \frac{5}{\sqrt{1-4x}} dx$

v $\int 16\sqrt[3]{6x+5} \, dx$

w $\int \left(5+(4x+1)^3\right) dx$

x $\int \frac{5}{\sqrt{x-2}} dx$

y $\int \left(\sqrt{x-2}+\sqrt{x+3}\right) dx$

2 Find the equation of y in terms of x, given that $\frac{dy}{dx} = 12(2x-1)^3$ and $y = 1$ when $x = 1$.

3 Find the equation of the curve that has a gradient function $\frac{dy}{dx} = 5(5x-6)^2$ and the point $(1, 1)$ lies on the curve.

4 For each of the following gradient functions, find the equation of y:

a $\frac{dy}{dx} = \sqrt{6x+66}$ and function y passes through the point $(-5, 17)$.

b $\frac{dy}{dx} = (4x-2)^3$ and function y passes through the point $(2, 79)$.

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Functions in the form f'(x).f(x)

5 Consider the function $\left(x^2-3\right)^5$.

a Calculate $\frac{d}{dx}\left(x^2-3\right)^5$.

b Hence find $\int 30x\left(x^2-3\right)^4\,dx$.

6 Consider the function $\left(5x^2+10x-3\right)^5$.

a Calculate $\frac{d}{dx}\left(5x^2+10x-3\right)^5$.

b Find $\int 150\left(x+1\right)\left(5x^2+10x-3\right)^4\,dx$.

7 Consider the function $\left(x^2-3x+6\right)^5$.

a Calculate $\frac{d}{dx}\left(x^2-3x+6\right)^5$.

b Hence find $\int \left(3-2x\right)\left(x^2-3x+6\right)^4\,dx$.

8 Consider the function $\sqrt{4x+11}$.

a Calculate $\frac{d}{dx}\left(\sqrt{4x+11}\right)$

b Hence find $\int \frac{12}{\sqrt{4x+11}}dx$.

9 Consider the function $\frac{1}{\left(x^2+7\right)^3}$.

a Calculate $\frac{d}{dx}\left(\frac{1}{\left(x^2+7\right)^3}\right)$.

b Hence find $\int -\frac{24x}{\left(x^2+7\right)^4}dx$.

10 Consider the function $y=e^{x^3}$.

a Calculate $\frac{dy}{dx}$.

b Hence find $\int 3x^2e^{x^3}\,dx$.

11 Consider the function $y=e^{3x^4-5}$ and hence find $36\int x^3e^{3x^4-5}dx$.

12 Consider the function $y=\sin\left(x^5\right)$ and hence find $\int x^4\cos\left(x^5\right)\,dx$.

13 Consider the function $y=\cos\left(x^6+3x^5\right)$ and hence find $15\int\left(2x^5+5x^4\right)\sin\left(x^6+3x^5\right)\,dx$.

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14 Consider the function $y = e^{\sqrt{x}}$ and hence find $\int \frac{4e^{\sqrt{x}}}{\sqrt{x}}\, dx$.

15 Find the following indefinite integrals:

a $\int 28\left(6x^3+1\right)\left(3x^4+2x\right)^6\, dx$

b $\int \sin 3x \cos 3x\, dx$

c $\int 2e^{\sin x}\cos x\, dx$

d $\int (x+2)e^{3x^2+12x}\, dx$

Integration as reverse of differentiation

16 Consider the function $y = \frac{\cos x}{e^x}$.

a Calculate $\frac{dy}{dx}$.

b Hence find $\int \frac{\sin x + \cos x}{e^x} dx$.

17 Consider the function $y = e^x \sin x$.

a Calculate $\frac{dy}{dx}$.

b Hence find $\int 3e^x(\sin x + \cos x)\, dx$.

18 Consider the function $y = \tan x$.

a Calculate $\frac{dy}{dx}$ using $\tan x = \frac{\sin x}{\cos x}$.

b Hence find $\int \frac{5}{\cos^2 x} dx$.

Rearrangement method

19 Given that $\sin 3t = 3\sin t - 4\sin^3(t)$

a Make $4\sin^3(t)$ the subject.

b Hence find $\int 3\sin^3(t)\, dt$.

20 Consider the function $y = xe^x$.

a Calculate $\frac{dy}{dx}$.

b Hence show that $xe^x = \int xe^x\, dx + \int e^x\, dx$.

c Rearrange the expression in part (b) to find $\int xe^x\, dx$.

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21 Consider the function $y = \dfrac{x}{e^x}$.

a Calculate $\dfrac{dy}{dx}$.

b Hence show that $\dfrac{x}{e^x} = \int \dfrac{1}{e^x}\,dx - \int \dfrac{x}{e^x}\,dx$.

c Rearrange the expression in part (b) to find $\int \dfrac{x}{e^x}\,dx$.

22 Consider the function $y = x\cos x$.

a Calculate $\dfrac{dy}{dx}$.

b Use the rearrangement method to find $\int x\sin x\,dx$.

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